► **Table 13.10** Data for parameter estimation exercise.

	docID	words in document	in $c = China$?
training set	1	Taipei Taiwan	yes
	2	Macao Taiwan Shanghai	yes
	3	Japan Sapporo	no
	4	Sapporo Osaka Taiwan	no
test set	5	Taiwan Taiwan Sapporo	?

tion about the test set was used in developing the classifier, the results of this experiment should be indicative of actual performance in practice.

This ideal often cannot be met; researchers tend to evaluate several systems on the same test set over a period of several years. But it is nevertheless highly important to not look at the test data and to run systems on it as sparingly as possible. Beginners often violate this rule, and their results lose validity because they have implicitly tuned their system to the test data simply by running many variant systems and keeping the tweaks to the system that worked best on the test set.

?

Exercise 13.6 [⋆⋆]

Assume a situation where every document in the test collection has been assigned exactly one class, and that a classifier also assigns exactly one class to each document. This setup is called one-of classification (Section 14.5, page 306). Show that in one-of classification (i) the total number of false positive decisions equals the total number of false negative decisions and (ii) microaveraged F_1 and accuracy are identical.

Exercise 13.7

The class priors in Figure 13.2 are computed as the fraction of *documents* in the class as opposed to the fraction of *tokens* in the class. Why?

Exercise 13.8

The function APPLYMULTINOMIALNB in Figure 13.2 has time complexity $\Theta(L_a + |\mathbb{C}|L_a)$. How would you modify the function so that its time complexity is $\Theta(L_a + |\mathbb{C}|M_a)$?

Exercise 13.9

Based on the data in Table 13.10, (i) estimate a multinomial Naive Bayes classifier, (ii) apply the classifier to the test document, (iii) estimate a Bernoulli NB classifier, (iv) apply the classifier to the test document. You need not estimate parameters that you don't need for classifying the test document.

Exercise 13.10

Your task is to classify words as English or not English. Words are generated by a source with the following distribution:

event	word	English?	probability
1	ozb	no	4/9
2	uzu	no	4/9
3	zoo	yes	1/18
4	bun	yes	1/18

(i) Compute the parameters (priors and conditionals) of a multinomial NB classifier that uses the letters b, n, o, u, and z as features. Assume a training set that reflects the probability distribution of the source perfectly. Make the same independence assumptions that are usually made for a multinomial classifier that uses terms as features for text classification. Compute parameters using smoothing, in which computed-zero probabilities are smoothed into probability 0.01, and computed-nonzero probabilities are untouched. (This simplistic smoothing may cause $P(A) + P(\overline{A}) > 1$. Solutions are not required to correct this.) (ii) How does the classifier classify the word zoo? (iii) Classify the word zoo using a multinomial classifier as in part (i), but do not make the assumption of positional independence. That is, estimate separate parameters for each position in a word. You only need to compute the parameters you need for classifying zoo.

Exercise 13.11

What are the values of $I(U_t; C_c)$ and $X^2(\mathbb{D}, t, c)$ if term and class are completely independent? What are the values if they are completely dependent?

Exercise 13.12

The feature selection method in Equation (13.16) is most appropriate for the Bernoulli model. Why? How could one modify it for the multinomial model?

Exercise 13.13

INFORMATION GAIN

Features can also be selected according to*information gain* (IG), which is defined as:

$$\text{IG}(\mathbb{D}, t, c) = H(p_{\mathbb{D}}) - \sum_{x \in \{\mathbb{D}_{t^+}, \mathbb{D}_{t^-}\}} \frac{|x|}{|\mathbb{D}|} H(p_x)$$

where H is entropy, $\mathbb{D}$ is the training set, and $\mathbb{D}_{t^+}$, and $\mathbb{D}_{t^-}$ are the subset of $\mathbb{D}$ with term t, and the subset of $\mathbb{D}$ without term t, respectively. p_A is the class distribution in (sub)collection A, e.g., $p_A(c) = 0.25$, $p_A(\overline{c}) = 0.75$ if a quarter of the documents in A are in class c.

Show that mutual information and information gain are equivalent.

Exercise 13.14

Show that the two X^2 formulas (Equations (13.18) and (13.19)) are equivalent.

Exercise 13.15

In the χ^2 example on page 276 we have $|N_{11} - E_{11}| = |N_{10} - E_{10}| = |N_{01} - E_{01}| = |N_{00} - E_{00}|$. Show that this holds in general.

Exercise 13.16

χ^2 and mutual information do not distinguish between positively and negatively correlated features. Because most good text classification features are positively correlated (i.e., they occur more often in c than in $\overline{c}$), one may want to explicitly rule out the selection of negative indicators. How would you do this?

13.7 References and further reading

General introductions to statistical classification and machine learning can be found in (Hastie et al. 2001), (Mitchell 1997), and (Duda et al. 2000), including many important methods (e.g., decision trees and boosting) that we do not cover. A comprehensive review of text classification methods and results is (Sebastiani 2002). Manning and Schütze (1999, Chapter 16) give an accessible introduction to text classification with coverage of decision trees, perceptrons and maximum entropy models. More information on the superlinear time complexity of learning methods that are more accurate than Naive Bayes can be found in (Perkins et al. 2003) and (Joachims 2006a).

Maron and Kuhns (1960) described one of the first NB text classifiers. Lewis (1998) focuses on the history of NB classification. Bernoulli and multinomial models and their accuracy for different collections are discussed by McCallum and Nigam (1998). Eyheramendy et al. (2003) present additional NB models. Domingos and Pazzani (1997), Friedman (1997), and Hand and Yu (2001) analyze why NB performs well although its probability estimates are poor. The first paper also discusses NB's optimality when the independence assumptions are true of the data. Pavlov et al. (2004) propose a modified document representation that partially addresses the inappropriateness of the independence assumptions. Bennett (2000) attributes the tendency of NB probability estimates to be close to either 0 or 1 to the effect of document length. Ng and Jordan (2001) show that NB is sometimes (although rarely) superior to discriminative methods because it more quickly reaches its optimal error rate. The basic NB model presented in this chapter can be tuned for better effectiveness (Rennie et al. 2003;Kołcz and Yih 2007). The problem of concept drift and other reasons why state-of-the-art classifiers do not always excel in practice are discussed by Forman (2006) and Hand (2006).

Early uses of mutual information and χ^2 for feature selection in text classification are Lewis and Ringuette (1994) and Schütze et al. (1995), respectively. Yang and Pedersen (1997) review feature selection methods and their impact on classification effectiveness. They find that *pointwise mutual information* is not competitive with other methods. Yang and Pedersen refer to expected mutual information (Equation (13.16)) as information gain (see Exercise 13.13, page 285). (Snedecor and Cochran 1989) is a good reference for the χ^2 test in statistics, including the Yates' correction for continuity for 2×2 tables. Dunning (1993) discusses problems of the χ^2 test when counts are small. Nongreedy feature selection techniques are described by Hastie et al. (2001). Cohen (1995) discusses the pitfalls of using multiple significance tests and methods to avoid them. Forman (2004) evaluates different methods for feature selection for multiple classifiers.

POINTWISE MUTUAL INFORMATION

David D. Lewis defines the ModApte split at www.daviddlewis.com/resources/testcollections/reuters215 based on Apté et al. (1994). Lewis (1995) describes *utility measures* for the

UTILITY MEASURE